

THE UNIVERSITY OF MELBOURNE

ELEN90098

Reinforcement Learning for Engineering

Workshop 1

Multi-Armed Bandits

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Question 1

Let $\mathcal{A}$ denote the set of available arms. For each arm $a \in \mathcal{A}$, its true action value is

$$q^*(a) = \mathbb{E}[R_k \mid A_k = a].$$

Let

$$a^* \in \arg \max_{a \in \mathcal{A}} q^*(a)$$

denote an optimal arm, with optimal expected reward

$$R^* = q^*(a^*) = \max_{a \in \mathcal{A}} q^*(a).$$

The cumulative expected regret after t time steps is

$$\text{Regret}_t = \mathbb{E} \left[\sum_{k=1}^t (R^* - R_k) \right]. \quad (1.1)$$

Equation (1.1) measures the expected reward lost relative to always selecting an optimal arm. The expectation accounts for both the randomness of the rewards and, where applicable, randomisation in the policy.

A bandit algorithm has *sublinear regret* if

$$\lim_{t \rightarrow \infty} \frac{\text{Regret}_t}{t} = 0, \quad (1.2)$$

or equivalently,

$$\text{Regret}_t = o(t).$$

Using (1.1),

$$\frac{\text{Regret}_t}{t} = R^* - \mathbb{E} \left[\frac{1}{t} \sum_{k=1}^t R_k \right]. \quad (1.3)$$

Combining (1.2) and (1.3) gives

$$\lim_{t \rightarrow \infty} \mathbb{E} \left[\frac{1}{t} \sum_{k=1}^t R_k \right] = R^*.$$

The significance of (1.2) is that the cost of learning becomes negligible relative to the length of the experiment. The cumulative regret may still increase with t , but it grows slowly enough that the regret incurred per time step vanishes. From (1.3), this means that the long-run average expected reward approaches the optimal expected reward R^* .

Equivalently, the algorithm can continue to explore and occasionally select suboptimal arms without sacrificing asymptotic performance, provided that the cumulative cost of these decisions grows sublinearly. For any strictly suboptimal arm, its fraction of selections must therefore vanish asymptotically. Hence, sublinear regret captures the desired balance between learning the unknown arm values and increasingly exploiting the optimal arm.

Question 2

To examine sublinear regret numerically, we estimate the cumulative pseudo-regret at a sequence of increasing horizons. Over M independent reruns,

$$\widehat{\text{Regret}}_T = \frac{1}{M} \sum_{r=1}^M \sum_{t=1}^T \left[R^* - q^*(A_t^{(r)}) \right]. \quad (2.1)$$

The true action values in (2.1) are used only for evaluation and are not available to the learning algorithm.

Three diagnostics are computed from this estimate. The first is the normalised regret

$$\hat{\rho}_T = \frac{\widehat{\text{Regret}}_T}{T}. \quad (2.2)$$

Since sublinear regret requires $\text{Regret}_T/T \rightarrow 0$, the trend $\hat{\rho}_T \rightarrow 0$ in (2.2) is the primary numerical criterion.

To measure the late-horizon growth directly, we also use the incremental regret rate

$$\hat{s}_i = \frac{\widehat{\text{Regret}}_{T_{i+1}} - \widehat{\text{Regret}}_{T_i}}{T_{i+1} - T_i}. \quad (2.3)$$

For geometrically spaced horizons $T_{i+1} = cT_i$,

$$0 \leq \hat{s}_i \leq \frac{\widehat{\text{Regret}}_{cT_i}}{(c-1)T_i} = \frac{c}{c-1} \frac{\widehat{\text{Regret}}_{cT_i}}{cT_i}.$$

Hence, sublinear regret implies that $\hat{s}_i \rightarrow 0$ in (2.3), whereas a positive late-horizon plateau is consistent with linear regret.

As a supplementary diagnostic, suppose locally that

$$\widehat{\text{Regret}}_T \approx CT^\alpha.$$

Then

$$\log \widehat{\text{Regret}}_T \approx \log C + \alpha \log T,$$

so

$$\hat{\alpha}_i = \frac{\log \widehat{\text{Regret}}_{T_{i+1}} - \log \widehat{\text{Regret}}_{T_i}}{\log T_{i+1} - \log T_i}. \quad (2.4)$$

Thus, $\hat{\alpha}_i < 1$ in (2.4) is consistent with locally sublinear power-law growth, while $\hat{\alpha}_i \approx 1$ indicates approximately linear growth. This diagnostic is supplementary because a sublinear function such as $T/\log T$ also has an effective exponent approaching 1.

The complete numerical procedure is given in Algorithm 1.

Algorithm 1 Multi-horizon numerical test for sublinear regret**Require:** Bandit policy π , horizons $T_1 < \dots < T_L$, number of reruns M

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1: for  $r = 1, \dots, M$  do
2:   Simulate  $\pi$  up to  $T_L$ 
3:   Record cumulative pseudo-regret  $C_{T_i}^{(r)}$  at each horizon  $T_i$ 
4: end for
5: for  $i = 1, \dots, L$  do
6:   Estimate  $\widehat{\text{Regret}}_{T_i} \leftarrow \frac{1}{M} \sum_{r=1}^M C_{T_i}^{(r)}$ 
7:   Compute normalised regret  $\widehat{\rho}_{T_i} \leftarrow \widehat{\text{Regret}}_{T_i} / T_i$ 
8: end for
9: for  $i = 1, \dots, L - 1$  do
10:  Compute incremental regret rate

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$$\widehat{s}_i \leftarrow \frac{\widehat{\text{Regret}}_{T_{i+1}} - \widehat{\text{Regret}}_{T_i}}{T_{i+1} - T_i}$$

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11:  Compute effective exponent

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$$\widehat{\alpha}_i \leftarrow \frac{\log \widehat{\text{Regret}}_{T_{i+1}} - \log \widehat{\text{Regret}}_{T_i}}{\log T_{i+1} - \log T_i}$$

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12: end for
13: Examine the late-horizon trends of  $\widehat{\rho}_{T_i}$  and  $\widehat{s}_i$ 
14: Use  $\widehat{\alpha}_i$  as a supplementary growth-rate diagnostic

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In our experiments, Algorithm 1 is applied at

$$T_i \in \{10^3, 3 \times 10^3, 10^4, 3 \times 10^4, 10^5\}.$$

Behaviour is considered consistent with sublinear regret when both $\widehat{\rho}_{T_i}$ and $\widehat{s}_i$ decrease toward zero at the largest horizons. Since only finitely many horizons and reruns are observed, this provides numerical evidence rather than a proof of asymptotic sublinearity.

Question 3

Q3: Sublinear regret of the ϵ -greedy schedules

The supplied ϵ -greedy implementation was extended using the multi-horizon test in Algorithm 1. Each schedule was simulated for $M = 500$ independent reruns up to $T = 10^5$, with the diagnostics from Question 2 evaluated at

$$T_i \in \{10^3, 3 \times 10^3, 10^4, 3 \times 10^4, 10^5\}.$$

The final exponential case in the supplied code was corrected to $\epsilon_t = 0.90^t$, consistent with its stated label.

The six schedules were

$$\epsilon_t \in \left\{ 0, 0.01, 0.10, \min \left[\left(\frac{k \log(t+1)}{t+1} \right)^{1/3}, 1 \right], 0.99^t, 0.90^t \right\}.$$

Figures 1–3 show the normalised regret, incremental regret rate, and effective exponent.

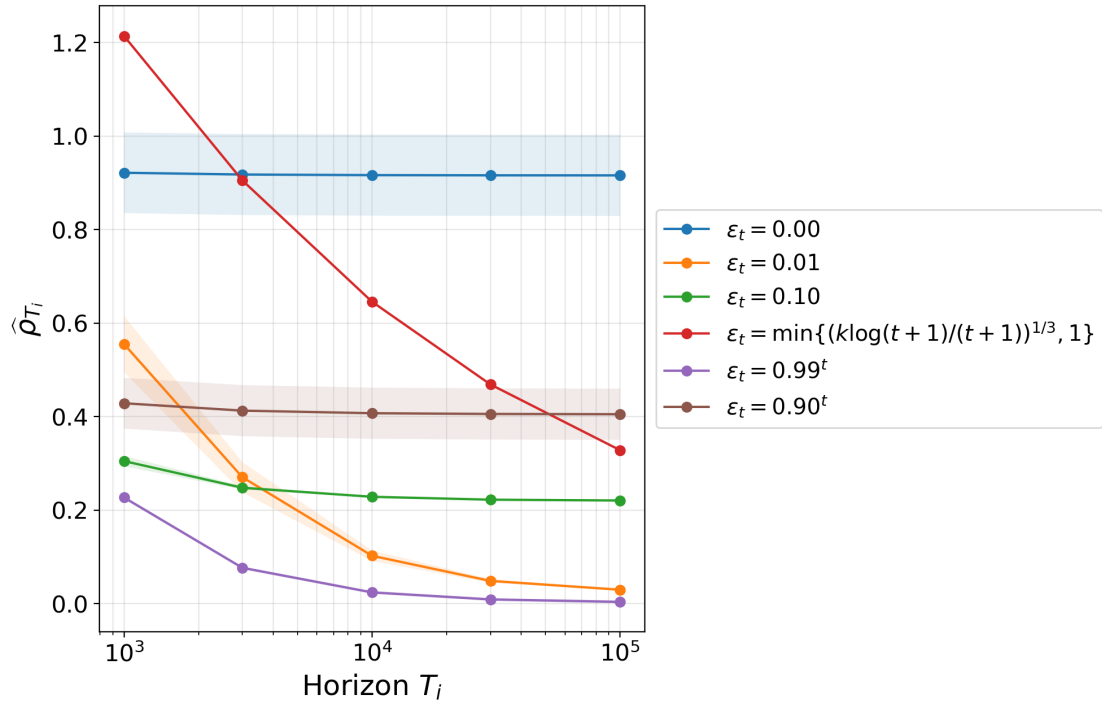


Figure 1: Normalised empirical cumulative pseudo-regret for the six ϵ -greedy schedules.

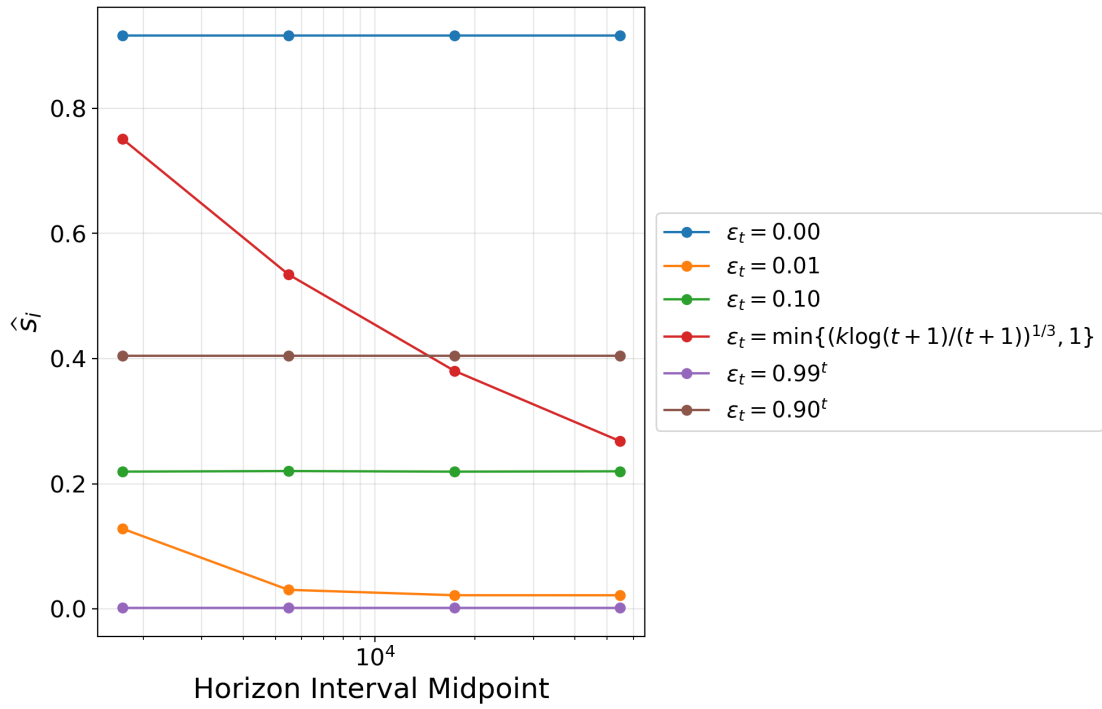


Figure 2: Incremental regret rate between consecutive simulation horizons.

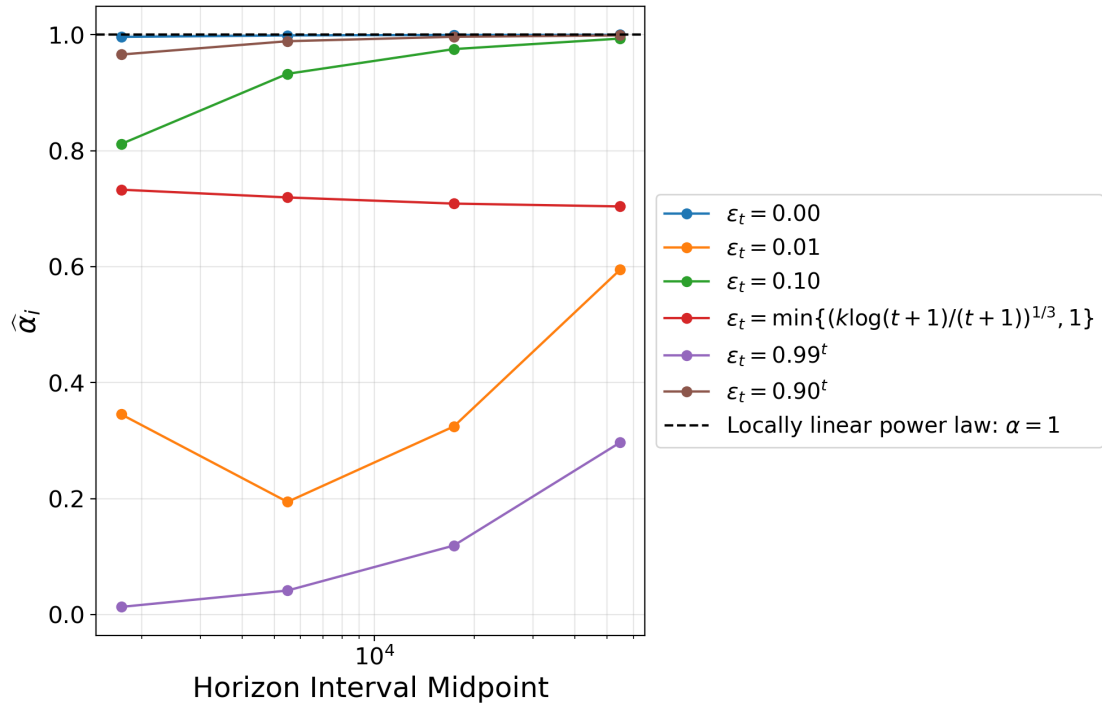


Figure 3: Effective regret exponent between consecutive horizons. The dashed line $\hat{\alpha} = 1$ corresponds to linear power-law growth.

The numerical results from Figures 1–3 are summarised in Table 1.

Table 1: Multi-horizon numerical diagnostics for the six ϵ -greedy schedules.

Schedule	$\hat{\rho}_{10^5}$	$\hat{s}_1 \rightarrow \hat{s}_4$	$\hat{\alpha}_1 \rightarrow \hat{\alpha}_4$
$\epsilon = 0$	0.9158	0.9157 $\rightarrow$ 0.9157 $\rightarrow$ 0.9157 $\rightarrow$ 0.9157	0.9963 $\rightarrow$ 0.9988 $\rightarrow$ 0.9996 $\rightarrow$ 0.9999
$\epsilon = 0.01$	0.0300	0.1282 $\rightarrow$ 0.0306 $\rightarrow$ 0.0220 $\rightarrow$ 0.0219	0.3456 $\rightarrow$ 0.1946 $\rightarrow$ 0.3246 $\rightarrow$ 0.5948
$\epsilon = 0.10$	0.2207	0.2194 $\rightarrow$ 0.2203 $\rightarrow$ 0.2194 $\rightarrow$ 0.2200	0.8115 $\rightarrow$ 0.9326 $\rightarrow$ 0.9752 $\rightarrow$ 0.9934
Polynomial decay	0.3283	0.7507 $\rightarrow$ 0.5345 $\rightarrow$ 0.3805 $\rightarrow$ 0.2681	0.7329 $\rightarrow$ 0.7195 $\rightarrow$ 0.7089 $\rightarrow$ 0.7040
$\epsilon_t = 0.99^t$	0.0040	0.001699 $\rightarrow$ 0.001697 $\rightarrow$ 0.001697 $\rightarrow$ 0.001697	0.0135 $\rightarrow$ 0.0417 $\rightarrow$ 0.1193 $\rightarrow$ 0.2970
$\epsilon_t = 0.90^t$	0.4052	0.4049 $\rightarrow$ 0.4049 $\rightarrow$ 0.4049 $\rightarrow$ 0.4049	0.9658 $\rightarrow$ 0.9888 $\rightarrow$ 0.9965 $\rightarrow$ 0.9989

The numerical diagnoses and their comparison with the theory covered in the lectures are summarised in Table 2.

Table 2: Numerical diagnosis of the six ϵ -greedy schedules and comparison with theoretical behaviour.

Schedule	Numerical diagnosis	Justification	Comparison with theory
$\epsilon = 0$	Not sublinear	$\hat{\rho}_T$ and $\hat{s}_i$ plateau near 0.916, while $\hat{\alpha}_i \rightarrow 1$.	Consistent: without exploration, an early incorrect estimate can cause permanent commitment to a suboptimal arm.
$\epsilon = 0.01$	Inconclusive	The late regret rate is small and $\hat{\rho}_T$ is still decreasing; the finite-horizon trends do not clearly establish the asymptotic regime.	Fixed $\epsilon > 0$ continues exploring suboptimal arms indefinitely, so theory predicts asymptotically linear regret.
$\epsilon = 0.10$	Not sublinear	$\hat{\rho}_T$ and $\hat{s}_i$ plateau at positive values, with $\hat{\alpha}_i \rightarrow 1$.	Consistent: persistent fixed exploration gives a non-zero asymptotic regret per step and hence linear cumulative regret.
Polynomial decay	Sublinear	Both $\hat{\rho}_T$ and $\hat{s}_i$ decrease, while $\hat{\alpha}_i$ falls from 0.733 to 0.704.	Strong agreement: the lecture gives $O(T^{2/3}(k \log T)^{1/3})$ regret.
$\epsilon_t = 0.99^t$	Inconclusive	Mean performance is excellent, but 499/500 runs have near-zero late slope while one has slope 0.84829.	Since $\sum_t 0.99^t < \infty$, only finite expected explicit exploration is available and rare permanent lock-in remains possible.
$\epsilon_t = 0.90^t$	Not sublinear	$\hat{\rho}_T$ and $\hat{s}_i$ plateau near 0.405, while $\hat{\alpha}_i \rightarrow 1$.	Consistent: exploration vanishes even faster and need not correct an early incorrect estimate.

For fixed $\epsilon > 0$, the theoretical linear-regret mechanism can be seen directly. Define the arm gap

$$\Delta_a = q^*(a^*) - q^*(a), \quad \bar{\Delta} = \frac{1}{k} \sum_{a \in \mathcal{A}} \Delta_a.$$

Once the optimal arm is known, the uniform exploration branch still incurs expected regret $\bar{\Delta}$ whenever it is used. Hence its contribution over T steps is

$$\epsilon \bar{\Delta} T = O(T).$$

This behaviour is already clear numerically for $\epsilon = 0.10$. For $\epsilon = 0.01$, the corresponding linear component is much smaller and the finite simulation does not resolve the asymptotic regime clearly enough; its numerical diagnosis is therefore retained as *inconclusive*.

The polynomial schedule gives the strongest validation of the theory. For

$$\epsilon_t = \min \left\{ t^{-1/3} (k \log t)^{1/3}, 1 \right\},$$

the analytical result gives

$$\text{Regret}_T = O\left(T^{2/3}(k \log T)^{1/3}\right). \quad (3.1)$$

The rate in (3.1) is sublinear because

$$\frac{T^{2/3}(k \log T)^{1/3}}{T} = \left(\frac{k \log T}{T}\right)^{1/3} \rightarrow 0.$$

Equation (3.1) also predicts the behaviour of the effective exponent. Ignoring the constant factor,

$$\begin{aligned} \log \text{Regret}_T &\approx \frac{2}{3} \log T + \frac{1}{3} \log \log T + C, \\ \alpha_{\text{th}}(T) &= \frac{d \log \text{Regret}_T}{d \log T} \approx \frac{2}{3} + \frac{1}{3 \log T} \rightarrow \frac{2}{3}. \end{aligned}$$

The measured exponent decreases from 0.733 to 0.704, closely following the theoretical trend towards $2/3$. Together with the decreasing normalised regret and incremental regret rate, this provides clear numerical evidence of sublinear regret for the polynomial schedule.

For the exponentially decaying schedules,

$$\sum_{t=0}^{\infty} 0.99^t = 100, \quad \sum_{t=0}^{\infty} 0.90^t = 10.$$

Thus, only a finite expected number of explicit exploratory actions occurs. If a suboptimal arm is preferred when exploration becomes negligible, that error may persist. This failure is already clear for 0.90^t . For 0.99^t , however, the mean hides a rare-event mixture: over the final interval $[30,000, 100,000]$, 499 of the 500 runs have essentially zero late slope, while one run has slope 0.84829. The finite-horizon evidence is therefore classified as *inconclusive* rather than sublinear.

Overall, the polynomial schedule is the only case with clear numerical evidence of sublinear regret. The cases $\epsilon = 0.01$ and $\epsilon_t = 0.99^t$ remain inconclusive at the simulated horizon, while $\epsilon = 0$, $\epsilon = 0.10$, and $\epsilon_t = 0.90^t$ exhibit clear non-sublinear behaviour.

Question 4

To test whether UCB learns each true action value $q^*(a)$, the algorithm was simulated for $T = 10^4$ steps over $M = 300$ independent reruns for $c \in \{0, 1, 2\}$. Each arm was selected once initially, after which the UCB index

$$Q_t(a) + c \sqrt{\frac{\log(t+1)}{N_t(a)}}$$

was used for action selection.

Since $q^*(a)$ is known in the synthetic testbed, the estimation error of arm a was measured by

$$\text{RMSE}_t(a) = \sqrt{\frac{1}{M} \sum_{r=1}^M \left(Q_t^{(r)}(a) - q^*(a) \right)^2}. \quad (4.1)$$

Thus, decreasing (4.1) indicates convergence of the empirical estimate towards the true arm value.

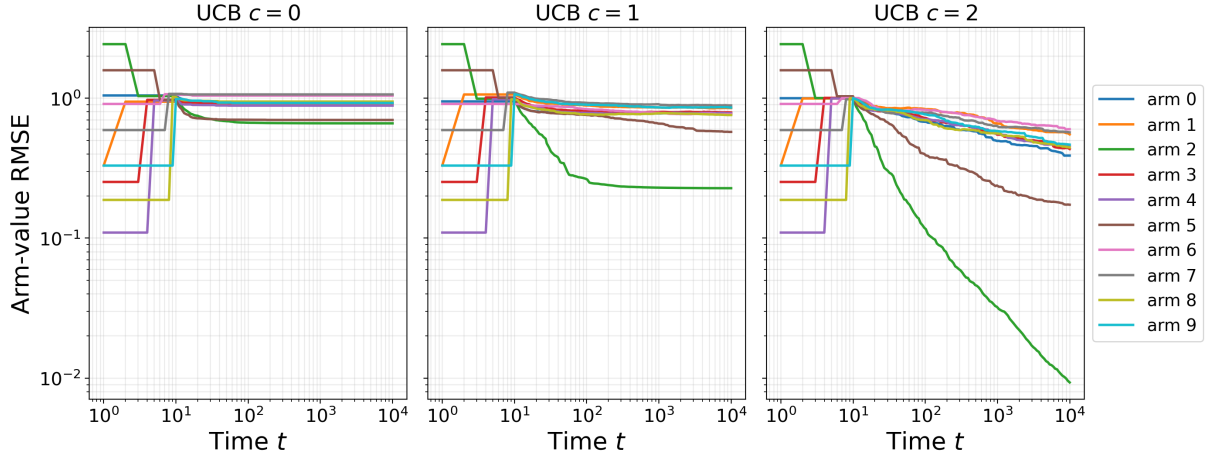


Figure 4: Arm-wise RMSE of the empirical action-value estimates under UCB for $c = 0, 1, 2$, averaged over $M = 300$ independent reruns.

The trends in Figure 4 are summarised at $T = 10^4$ in Table 3.

Table 3: Final arm-value estimation accuracy at $T = 10^4$.

c	Mean RMSE	Observation
0	0.8981	Many arms are not revisited.
1	0.7254	Improved, but poor arms remain sparsely sampled.
2	0.4090	Best estimation across all arms.

Figure 4 shows that stronger exploration improves estimation of the full set of arm values. With $c = 0$, UCB becomes greedy after initialisation, so arms that appear suboptimal may never be sampled again and their estimation errors remain large. For $c = 1$ and $c = 2$, the confidence bonus causes uncertain arms to be revisited, with $c = 2$ giving the lowest mean arm-wise RMSE in this experiment.

This behaviour follows directly from the UCB index. For any $c > 0$, if $N_t(a)$ remained bounded for some arm, then

$$c \sqrt{\frac{\log(t+1)}{N_t(a)}} \rightarrow \infty,$$

eventually forcing that arm to be selected again. Hence exploratory UCB continues to collect samples from every arm, allowing the sample-average estimate $Q_t(a)$ to approach $q^*(a)$. The numerical results therefore support eventual learning for $c > 0$, although convergence of highly suboptimal arms can be slow at finite horizons. The case $c = 0$ has no such exploration mechanism and does not generally learn every arm.

Question 5

The UCB and ϵ -greedy algorithms were compared on the same fixed 10-armed bandit over a horizon of $T = 10^4$, with $M = 300$ independent reruns. For each run, cumulative pseudo-regret was computed as

$$C_T = \sum_{t=1}^T [R^* - q^*(A_t)], \quad (5.1)$$

and then averaged across reruns. Equation (5.1) was evaluated for UCB with $c \in \{0, 1, 2\}$ and for the six ϵ -greedy schedules considered in Question 3.

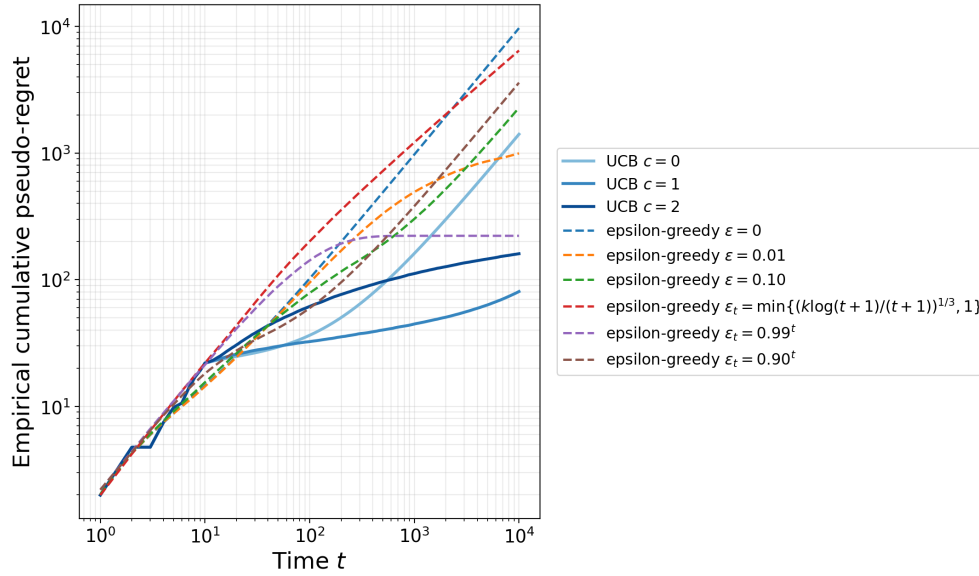


Figure 5: Empirical cumulative pseudo-regret of UCB and ϵ -greedy policies over $T = 10^4$ time steps.

The main results from Figure 5 are summarised in Table 4.

Table 4: Empirical cumulative pseudo-regret at $T = 10^4$.

Policy	Regret	Observation
UCB $c = 1$	80.58	Lowest regret.
UCB $c = 2$	160.29	More exploration; better value estimates but higher regret.
$\epsilon_t = 0.99^t$	221.98	Best ϵ -greedy result at this horizon.
$\epsilon = 0.01$	994.19	Persistent random exploration increases regret.
UCB $c = 0$	1408.07	No systematic exploration; susceptible to incorrect commitment.
$\epsilon = 0.10$	2278.42	Frequent random exploration produces large regret.
Polynomial decay	6444.72	Still explores aggressively at this finite horizon.

Figure 5 and Table 4 show that the exploratory UCB policies achieve substantially lower finite-horizon regret than the tested ϵ -greedy schedules. UCB with $c = 1$ performs best, with regret 80.58, followed by $c = 2$ with regret 160.29. The best-performing ϵ -greedy schedule is $\epsilon_t = 0.99^t$, with regret 221.98.

The comparison with Question 4 reveals an important distinction between learning the arm values and minimising regret. In Question 4, UCB with $c = 2$ achieved the lowest mean arm-wise RMSE and therefore learned the complete set of action values most accurately. However, here $c = 1$ achieves the lower regret. Increasing c encourages more exploration, which improves the estimates of uncertain and suboptimal arms, but selecting those arms also incurs regret. Thus,

better estimation of all arm values does not necessarily produce better cumulative reward.

This directly illustrates the exploration–exploitation trade-off. UCB with $c = 1$ explores sufficiently to identify the high-value arms while avoiding some of the additional exploratory cost incurred by $c = 2$. In contrast, $c = 0$ moves too far towards exploitation: without systematic exploration, an early estimation error can cause persistent selection of a suboptimal arm.

The ϵ -greedy results show a similar trade-off, but exploration is undirected. Fixed $\epsilon > 0$ continues selecting random arms even after the optimal arm has largely been identified, producing substantial cumulative regret. The polynomially decaying schedule performs particularly poorly over $T = 10^4$ because it is still exploring aggressively at this horizon. This does not contradict the sublinear long-run behaviour identified in Question 3; it demonstrates that favourable asymptotic regret growth does not imply good finite-horizon performance.

Therefore, for this bandit instance and horizon, UCB with $c = 1$ provides the best exploration–exploitation balance and achieves the lowest cumulative pseudo-regret. Together with Question 4, the results show that the amount of exploration that is best for learning the full reward model need not be the amount that is best for maximising reward.

Question 6

The gradient-bandit and UCB algorithms were compared on the same fixed 10-armed bandit over $T = 10^4$ time steps and $M = 300$ independent reruns. The comparison used UCB with $c \in \{0, 1, 2\}$ and gradient bandit with $\alpha \in \{0, 0.1, 0.5\}$.

Unlike UCB, the gradient-bandit algorithm does not estimate $q^*(a)$ directly. Instead, it learns action preferences $H_t(a)$ and forms the softmax policy

$$\pi_t(a) = \frac{\exp(H_t(a))}{\sum_b \exp(H_t(b))}.$$

Using the running average reward $\bar{R}_t$ as the baseline, the preference update can be written as

$$H_{t+1}(a) = H_t(a) + \alpha(R_t - \bar{R}_t) [\mathbf{1}_{\{a=A_t\}} - \pi_t(a)].$$

Thus, α directly controls how strongly each observed reward changes the policy.

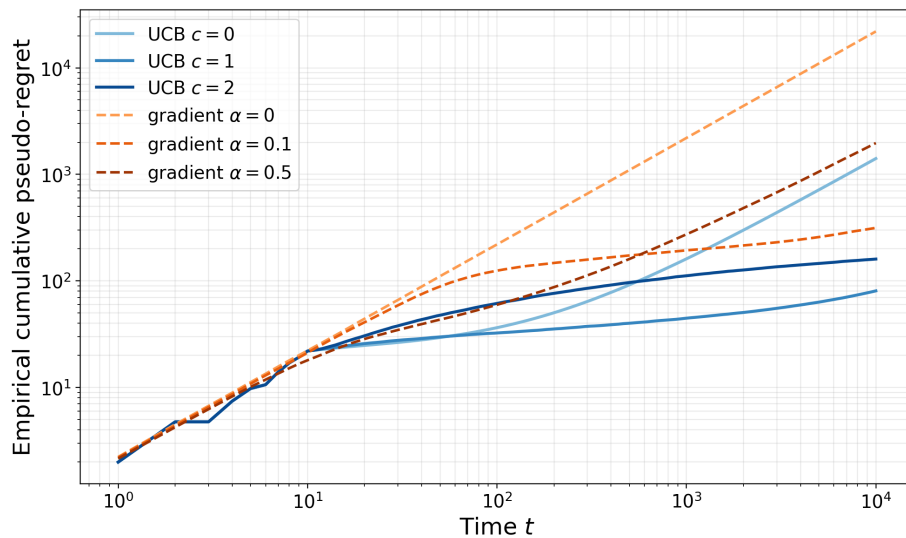


Figure 6: Empirical cumulative pseudo-regret of the gradient-bandit and UCB algorithms over $T = 10^4$ time steps.

The final cumulative pseudo-regrets from Figure 6 are summarised in Table 5.

Table 5: Empirical cumulative pseudo-regret at $T = 10^4$.

Policy	Regret	Interpretation
UCB, $c = 1$	80.58	Lowest regret overall.
UCB, $c = 2$	160.29	More exploration than $c = 1$, but still low regret.
Gradient, $\alpha = 0.1$	314.58	Best gradient case; moderate preference updates.
UCB, $c = 0$	1408.07	No systematic exploration.
Gradient, $\alpha = 0.5$	1968.16	Aggressive updates produce substantially larger regret.
Gradient, $\alpha = 0$	21978.00	No learning; the policy remains uniform.

The three gradient-bandit cases show a strong dependence on the step size α . For $\alpha = 0$, the preferences never change:

$$H_t(a) = H_0(a) = 0 \implies \pi_t(a) = \frac{1}{k}.$$

The policy therefore selects all arms uniformly for the entire experiment. Because suboptimal arms continue to be selected with fixed probability, the regret grows approximately linearly, giving the very large final value 21978.

With $\alpha = 0.1$, the preference updates are large enough to learn useful differences between the arms without reacting too aggressively to individual noisy rewards. This is the best gradient-bandit configuration tested, with cumulative regret 314.58. However, it still accumulates

$$\frac{314.58}{80.58} \approx 3.90$$

times the regret of UCB with $c = 1$.

Increasing the step size to $\alpha = 0.5$ makes each reward observation have a much larger effect on the preferences. Since rewards are noisy, larger updates can produce excessive preference changes and cause the softmax policy to concentrate too strongly on arms favoured by early observations. Its cumulative regret rises to 1968.16, more than six times that of the $\alpha = 0.1$ case.

Overall, the tested gradient-bandit parameters exhibit the expected step-size trade-off:

$\alpha = 0$: no learning, $\alpha = 0.1$: effective learning, $\alpha = 0.5$: overly aggressive learning.

Among the tested policies, exploratory UCB is substantially more regret-efficient: UCB with $c = 1$ achieves the lowest regret, while the best gradient-bandit case incurs approximately 3.90 times more regret. This is an empirical result for the present stationary Gaussian testbed and finite horizon, rather than a general claim that UCB always outperforms gradient bandit.

Question 7

To examine the sensitivity of gradient bandit to the step size, the experiment was repeated for

$$\alpha \in \{0.01, 0.05, 0.10, 0.50, 1, 2, 5\}$$

over $T = 10^4$ time steps and $M = 300$ independent reruns. For each α , we monitored the probability assigned to the true optimal arm, $\pi_t(a^*)$, and the cumulative pseudo-regret.

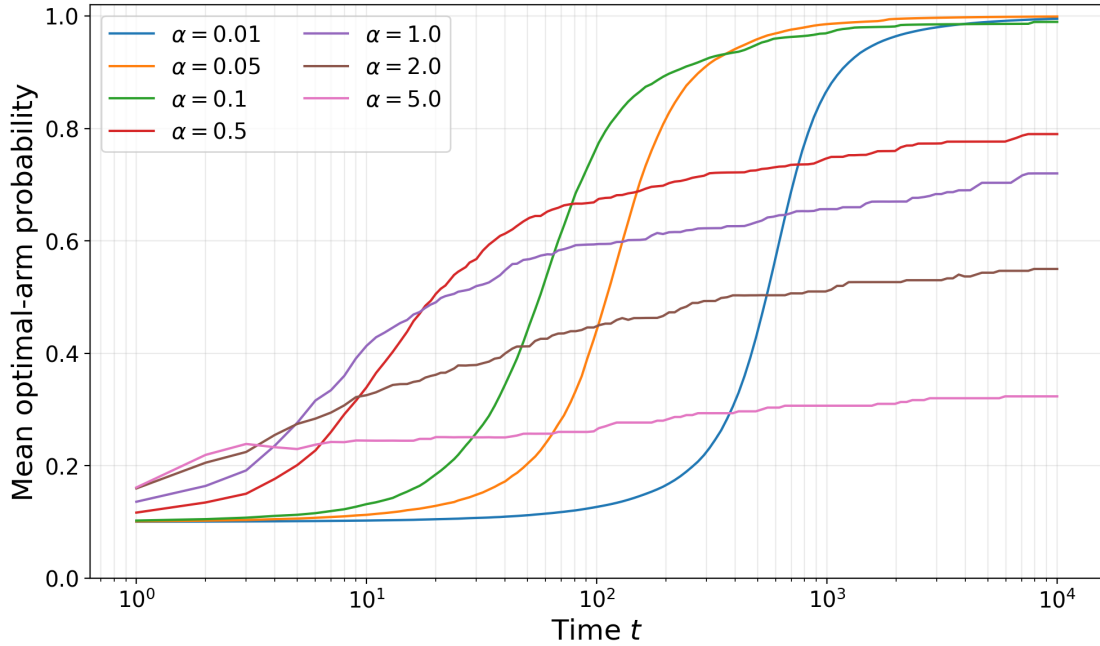


Figure 7: Mean probability of selecting the optimal arm for different gradient-bandit step sizes.

The final performance in Figure 7 is summarised in Table 6.

Table 6: Gradient-bandit performance at $T = 10^4$.

α	Mean $\pi_T(a^*)$	Regret	Interpretation
0.01	0.9952	1542.45	Reliable but slow learning.
0.05	0.9991	357.39	Strong finite-horizon performance.
0.10	0.9896	314.58	Lowest regret.
0.50	0.7899	1968.16	Frequent premature lock-in.
1.00	0.7200	2765.23	Increasingly unreliable.
2.00	0.5500	5358.04	Strong sensitivity to early rewards.
5.00	0.3233	11148.71	Most runs lock onto a suboptimal arm.

The results reveal three distinct step-size regimes. For very small α , learning is stable but slow. For example, $\alpha = 0.01$ eventually gives $\pi_T(a^*) = 0.9952$, yet accumulates 1542.45 regret because the policy takes a long time to concentrate on the optimal arm.

Moderate step sizes give the best finite-horizon compromise. Both $\alpha = 0.05$ and $\alpha = 0.10$ concentrate strongly on the optimal arm while incurring much less regret. In particular, $\alpha = 0.10$ gives the lowest regret, 314.58, among the tested gradient-bandit configurations.

For large α , performance deteriorates rapidly. The preference update contains the factor

$$\alpha(R_t - \bar{R}_t),$$

so a larger α amplifies the effect of individual noisy rewards. An early favourable reward can therefore produce a large preference shift, which softmax converts into a highly concentrated

policy:

early reward noise $\longrightarrow$ large preference shift $\longrightarrow$ policy concentration
 $\longrightarrow$ reduced opportunity to correct an early error.

The mean curves alone hide an important run-level effect. At $\alpha = 5$, the final policies are strongly bimodal, as shown in Table 7.

Table 7: Run-level outcome at $\alpha = 5$ over 300 reruns.

Final policy condition	Fraction of runs
$\pi_T(a^*) > 0.95$	32.33%
$\pi_T(a^*) < 0.05$	67.67%

Thus, the mean value $\pi_T(a^*) = 0.3233$ at $\alpha = 5$ does not represent a typical moderately uncertain policy. Instead, individual runs almost completely commit either to the optimal arm or to a suboptimal arm, with most runs in the latter category. The deterioration in the mean optimal-arm probability as α increases is therefore caused by an increasing frequency of premature lock-in, rather than by all runs becoming gradually less confident.

Overall, the numerical results do not support reliable convergence for arbitrary $\alpha > 0$. Very small step sizes learn reliably but slowly, moderate values around 0.05–0.10 give the best finite-horizon performance, while large step sizes can amplify reward noise and cause near-deterministic commitment to a suboptimal arm. Since the experiment is finite, this provides numerical evidence of strong step-size sensitivity rather than a general asymptotic non-convergence theorem.